Aditya S

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Thiruvananthapuram, Kerala, India | Open to remote

SUMMARY

Product engineer at UST, building web products backed by AWS Bedrock. Founder and lead maintainer of open-source developer tools used by over 1.8 million people across 25,000+ deployments, and a member of Anthropic's Claude for Open Source Program. Works across the whole stack: Angular front end, Spring Boot APIs, and the retrieval and guardrail layer behind the model.

EXPERIENCE

Associate 2, Product Engineering **Jul 2025 to Present**
UST *Thiruvananthapuram, India*

- Promoted from Associate 1 to Associate 2 in August 2026.
- Build and maintain sales and operations features on an enterprise telecom platform, working across the Angular front end and the Spring Boot REST services behind it.
- Led the front-end migration from **Angular 15 to Angular 21**, rewriting a large share of the components and shared modules to fit standalone components, the new control flow, and signals.
- Deliver generative-AI features on AWS: retrieval pipelines on **Amazon Bedrock Managed Knowledge Bases**, policy enforcement through **Bedrock Guardrails**, and product features backed by Claude Sonnet.
- Cut API response times by up to **40%** by restructuring REST endpoints, tightening DTO boundaries, and rewriting inefficient database queries.
- Produce product demo films in Remotion (React), scripted and rendered from code, for expo booths and client briefings.

Tech Intern **2024**
BreadcrumbsAI *Remote*

- Built Python scraping services with Playwright for dynamic extraction and BeautifulSoup for parsing. Added structured logging so long-running jobs failed loudly instead of dropping records silently.

Project Lead & Backend Developer **2022 to 2024**
Google Developer Student Clubs, MBCET *Kerala, India*

- Led club technical projects; built the GDSC MBCET website backend and event automation tooling.

Campus Lead **2023 to 2024**
GTECH µLearn, MBCET *Kerala, India*

- Ran campus learning and skill-development programmes; campus crossed 1 million karma points.

OPEN SOURCE

Ultroid: Telegram Userbot Framework *Founder & Lead Maintainer, 2021 to Present*

- Founded and lead a Python automation framework with **3k+ GitHub stars** and **25,000+ deployed instances** worldwide.
- Designed the plugin architecture that carries 200+ modules, so contributors can extend the runtime without touching core.
- Run releases, code review, and issue triage for a distributed contributor base. github.com/TeamUltroid/Ultroid

Stack: Python, MongoDB, SQLite, Docker, Shell

ChannelActions: Telegram Bot *Creator & Maintainer*

- Automates Telegram channel join requests (auto-approval, auto-decline, custom welcome flows) for **1.8M+ active users**, localised into 25+ languages.
- Built the broadcast system that delivers to that user base: async Python with `motor`, token-bucket rate limiting to stay inside API quotas, SQLite for resumable delivery state, and `_id`-based cursor pagination over MongoDB Atlas.
- Rewrote the service on Deno with Deno KV, and shipped a web dashboard so admins configure it themselves.
- channelactions.xditya.me | github.com/xditya/ChannelActionsBot

Stack: TypeScript, Deno, Deno KV, Python, MongoDB Atlas, SQLite, i18n

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C#, Java, Kotlin, Dart, C++, SQL
Frontend	Angular, React, Next.js, Vite, Remotion
Mobile & Desktop	React Native, Expo, Flutter, Jetpack Compose, Swift and Kotlin native modules, .NET 10, WinUI 3, Win32
Backend & Data	Spring Boot, FastAPI, Node.js, Deno, REST API design, MongoDB Atlas, SQLite, Deno KV, Vercel KV
Cloud & AI	AWS, Amazon Bedrock (Knowledge Bases, Guardrails, Claude), RAG pipelines, MCP, Gemini API, Agno
Tooling	Git, GitLab CI/CD, GitHub Actions, Docker, ELK, Vercel, ffmpeg, LaTeX

SELECTED PROJECTS

engram: Local-First Knowledge Library *Android & iOS, AGPL-3.0 | Aug 2026 to Present*

- Open-source app for saving links, articles, images and notes, with no account and no server. Every device holds a full SQLite copy, and everything exports in one tap to files, JSON, CSV, or Obsidian markdown.
- Built **end-to-end encrypted sync** that runs through the user's own Google Drive, iCloud or WebDAV: an append-only op-log with per-field last-writer-wins over hybrid logical clocks, content-addressed blobs, XChaCha20-Poly1305, and BIP39 recovery phrases. Property and fuzz tested.
- One-tap capture from any app, using a translucent overlay activity on Android and an iOS share extension that boots a slim React Native root inside the extension process.
- Tagging and full-text search work with no model at all, over SQLite FTS5 with `tag:`, `site:` and `before:` operators. An optional intelligence layer adds summaries and visual search through a bring-your-own API key or on-device inference (ExecuTorch, Qwen3 0.6B with MiniLM embeddings). engram.xditya.me | github.com/xditya/engram

Stack: TypeScript monorepo (pnpm), React Native, Expo SDK 57, op-sqlite, Swift and Kotlin native modules, GitHub Actions CI

Tomte: Windows 11 Utility *C# / .NET 10, MIT | Microsoft Store, Oct 2026*

- Open-source utility that stays hidden until called, then grows out of the top edge of the screen as a pill and panels. Holds a file shelf, app and script shortcuts, media controls, a Markdown scratchpad, clipboard history, timers, and search across apps, files and Settings pages.
- Drew the island as a focus-free Win32 window with **Windows.UI.Composition**, kept out of the taskbar and Alt+Tab, with WinUI 3 panels hosted through `DesktopWindowXamlSource`, spring-animated shape changes, and input hooks on a dedicated thread.
- Optional AI layer runs on the user's own key or locally through Ollama, with MCP tool support. Nothing reaches a provider without a visible user action, and the UI shows where each request went and what it cost.
- Shipped it end to end: **~1,870 unit tests**, GitHub Actions for build, test and format checks, MSIX packaging and Store listing, winget install, and the product site on Cloudflare Workers. tomte.xditya.me | github.com/xditya/tomte-windows

Stack: C#, .NET 10, Win32 interop (CsWin32), WinUI 3, SQLite, MSIX, GitHub Actions, Cloudflare Workers

Splitty: Bill-Splitting PWA *React, TypeScript, Vite*

- Splits restaurant bills from a photo. OCR falls back through three tiers: a user-supplied Gemini key, then a shared server-side proxy, then `tesseract.js` as WASM in the browser, so the app degrades instead of failing.
- Wrote an integer-paise allocator using the largest-remainder method, fuzz-tested so splits reconcile exactly to the bill total with no rounding drift.
- Handles GST, liquor VAT at its own rate, and service charge applied before tax. Settle-up runs through UPI QR codes, and sessions are shareable via Vercel KV. splitty.xditya.me | github.com/xditya/Splitty

EDUCATION

B.Tech, Electronics and Communication Engineering **2021 to 2025**
Mar Baselios College of Engineering and Technology *Kerala, India*

CERTIFICATIONS & ACHIEVEMENTS

- **AWS:** Amazon Bedrock Managed Knowledge Bases; Amazon Bedrock Guardrails.
- **Anthropic:** Claude for Open Source Program member, and Claude Code Partner Badge.
- **Winner, EDEN 3.0 Hackathon** (Marian Engineering College, 2024) for an adaptive employee-management platform with automated promotion calculation.
- **Top 100 Coders, Kerala** (μLearn at HuddleGlobal, 2023), recognised by the Government of Kerala.
- **#12 in India** for public GitHub contributions, 2021; 660+ followers on GitHub.